

Assignment 1: Digital Image Processing

1. If an image has dimensions of 128x128 pixels with 3 bands and each pixel is represented by 8 bits, what is the total memory size required to store this image? How does the required memory size change if the image is converted to grayscale?
2. An image segment is shown below. Let, V be the set of gray level values used to define connectivity in the image. Show all kinds of connectivity for (i) $V = \{2,3\}$ (ii) $V = \{2,6\}$. Compute D_e , D_4 , D_8 distances between pixels 'p' and 'q'.

2 (p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	6 (q)

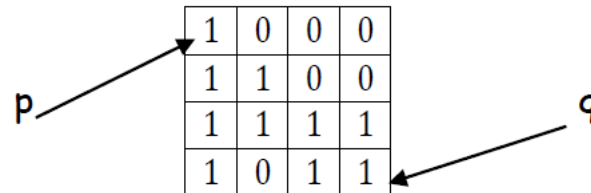
3. What are the various steps in digital image processing? Draw its flow diagram.
4. Generally, a transmission is accomplished in the form of packets consisting of a start bit, a byte of information and a stop bit. Considering this fact, answer the following:
 - (a) For an image of 1024x1024 with 256 gray levels how much time will it take over 750K baud modem?
 - (b) How many minutes would it take transmit a 1024x1024 image with 256 gray levels if we use a 56k baud modem? if 1 bit is used for start and end data.
5. Consider an image segment given below.

3	4	1	2	0
0	1	0	4	Q: 2
2	2	3	1	4
P: 3	0	4	2	1

- a. Let $V = \{0, 1, 2\}$, compute the length of the shortest 4-, 8-, and m-path between p and q
 - b. Repeat for $V = \{2, 3, 4\}$
6. Suppose there is a binary image of 100*100. How many distinct binary images can be generated?
 7. An image scan center needs to store CAD images whose resolution is 1024x1024, with 64 gray levels. A total of 100 layers are present.
 - A. How much storage is required in GB.
 - B. Suppose this image is transferred via medium having a transfer rate of 128 K bits/second in the form of packets. The size of a packet is 100 bits and each packet is segregated by a

start and a stop flag of 14 bits each. Compute the time required to transfer the image (in hrs).

8. Define various types of image resolution. One mobile phone has higher spatial resolution while another has higher gray level. Which mobile would you prefer to buy and why?
9. Mark the m-adjacent path from p to q in the following image.



10. The distance between pixels p & q in the above image:
 - a. Euclidean distance:
 - b. City block distance:
 - c. Chess board distance: